

DUAN Zihang Raymond

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Computer Engineering student passionate about AI, IoT, and Full-Stack Development.

Education

Hong Kong University of Science and Technology (HKUST) Sep 2026 Intake
MSc in Information Technology
Hong Kong University of Science and Technology (HKUST) Sep 2022 – Jun 2026
BEng in Computer Engineering — Second Class Honors, Division 1
Coursework: AI & Vision, Systems & Networks, Software Engineering, Architecture & Multimedia

Work Experience

Growgreen Limited Jun 2026 – Jul 2026
Software Engineer Intern Science Park, Hong Kong

- **Product Analysis:** Disassembled and studied company products to understand hardware design, component architecture, and integration requirements.
- **Website Migration (Team Lead):** Led intern team to migrate the company website from WordPress to Shopify; built and customized storefront pages using Liquid and CSS.

Intelli Global Corporation Jun 2025 – May 2026
QA Engineer Intern (Part Time and Full Time) Kwun Tong, Hong Kong

- **Data Integrity & Automation:** Engineered Python (NumPy) scripts to transform, clean, and validate large-scale datasets and raw CSV files, ensuring seamless integration into the project database.
- **Quality Assurance:** Identified, reported, and verified bug fixes; conducted rigorous regression testing cycles to ensure system stability following new feature deployments.
- **Documentation:** Managed system specifications and user guides; standardized technical manuals to improve clarity and reduce support queries.
- **Strategic Collaboration:** Partnered with Project Managers to provide UI/UX improvement suggestions, influencing the product roadmap for smart residential solutions.
- **System Deployment:** Deployed and configured openMAINT using Docker on Windows, streamlining asset management for smart building data.

Featured Projects

MTR Tracker Lite — *React Native, Expo, REST APIs* — Built a streamlined mobile app for real-time Hong Kong MTR tracking; integrated GPS, train APIs, and HKO weather data for a fast, ad-free experience.

Muse Dash Web Game — *Node.js, Express, Socket.io, Bcrypt* — Built a real-time backend with bidirectional event-based communication; implemented secure user authentication and password hashing.

Library Management System — *OOP, UML, Java* — Designed software architecture with use-case specifications for user roles, publishing pipelines, access control, and status tracking dashboards.

Fruit Ripeness Classification — *Python, Transfer Learning, EfficientNet, YOLO* — Trained an EfficientNet-B0 model to classify ripeness stages of three fruit types; used YOLO bounding boxes to segment inputs, achieving 93% test accuracy.

Tiny Smart Home — *ESP32, Home Assistant, IoT, Ollama AI* — Built a smart home system with ESP32 sensors (DHT11, MQ2, UV); visualized data on a dashboard, built a physical demo model, and implemented automation scripts.

Technical Skills

Languages: Python, C/C++, JavaScript/Node.js, HTML/CSS, Java
Spoken: Cantonese (4.5/5), English (4/5), Mandarin (4.5/5)
AI & ML: PyTorch, NumPy & Scikit-learn, Deep Learning (CNN, RNN), Computer Vision, GAN & Transformers, Reinforcement Learning
Backend & Systems: MySQL, Express.js & Socket.io, Linux, Object-Oriented Design
Tools & Hardware: Git, STM32/ESP32, MIPS Architecture, Hardware Communication Protocols (I2C, ADC), Docker